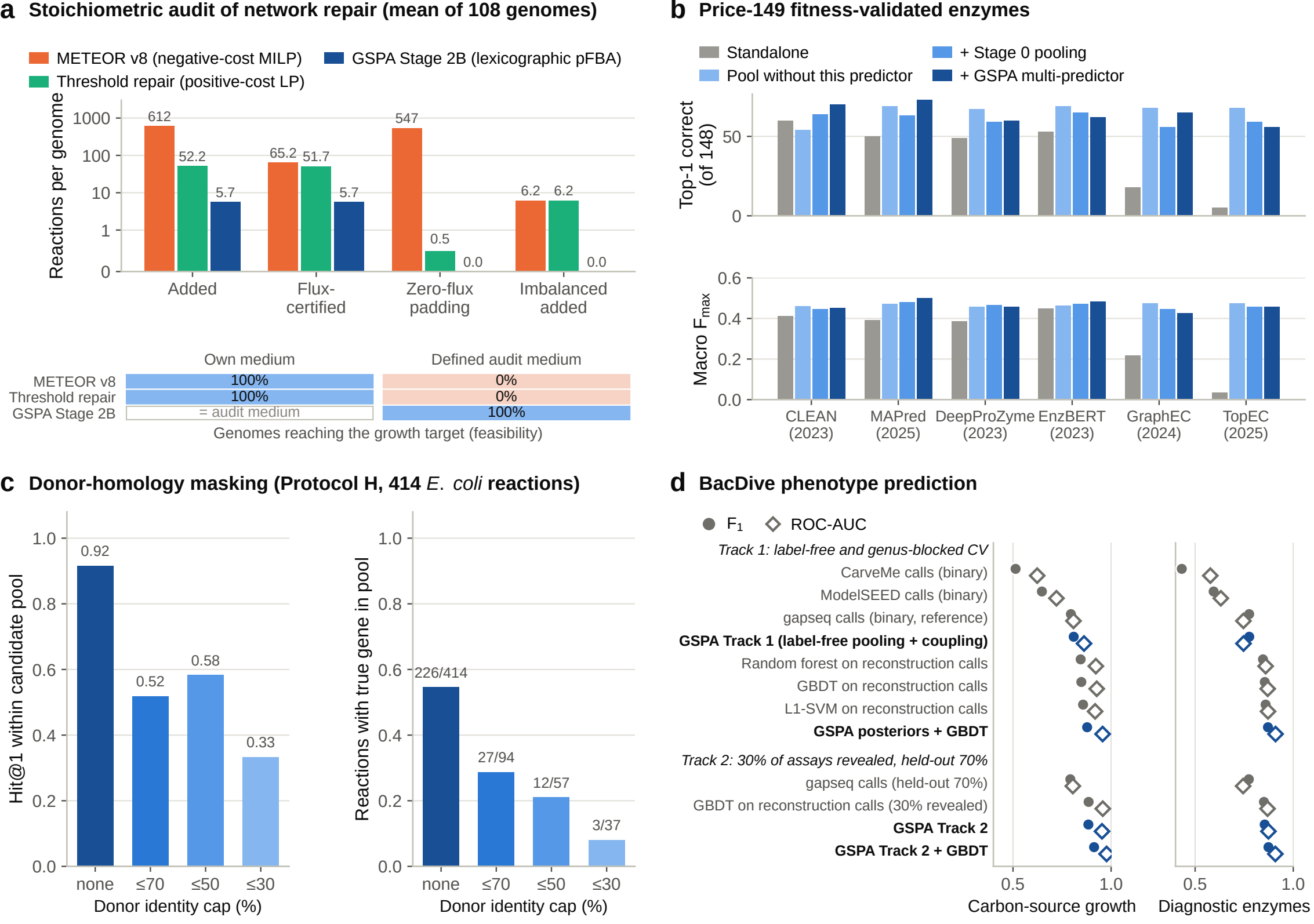
a Stoichiometric audit of network repair (mean of 108 genomes)
METEOR v8 (negative-cost MILP)
GSPA Stage 2B (lexicographic pFBA)
Threshold repair (positive-cost LP)
Reactions per genome
1000
100
10
1
0
612
52.2
5.7
65.2
51.7
5.7
547
0.5
0.0
6.2
6.2
0.0
Added
Flux-certified
Zero-flux padding
Imbalanced added
Own medium
Defined audit medium
METEOR v8
Threshold repair
GSPA Stage 2B
100%
100%
= audit medium
0%
0%
100%
Genomes reaching the growth target (feasibility)
b Price-149 fitness-validated enzymes
Standalone
Pool without this predictor
+ Stage 0 pooling
+ GSPA multi-predictor
Top-1 correct (of 148)
50
0
Macro F_{max}
0.6
0.4
0.2
0.0
CLEAN (2023)
MAPred (2025)
DeepProZyme (2023)
EnzBERT (2023)
GraphEC (2024)
TopEC (2025)
c Donor-homology masking (Protocol H, 414 E. coli reactions)
Hit@1 within candidate pool
1.0
0.8
0.6
0.4
0.2
0.0
0.92
0.52
0.58
0.33
none
≤70
≤50
≤30
Donor identity cap (%)
Reactions with true gene in pool
226/414
27/94
12/57
3/37
d BacDive phenotype prediction
F_1
ROC-AUC
Track 1: label-free and genus-blocked CV
CarveMe calls (binary)
ModelSEED calls (binary)
gapseq calls (binary, reference)
GSPA Track 1 (label-free pooling + coupling)
Random forest on reconstruction calls
GBDT on reconstruction calls
L1-SVM on reconstruction calls
GSPA posteriors + GBDT
Track 2: 30% of assays revealed, held-out 70%
gapseq calls (held-out 70%)
GBDT on reconstruction calls (30% revealed)
GSPA Track 2
GSPA Track 2 + GBDT
0.5
1.0
Carbon-source growth
Diagnostic enzymes